Computational Analysis of Sealed-Bid Auctions Using Auction Theory

A simulation-based study of first-price and second-price auction mechanisms

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Abstract

English

This research investigates the strategic and computational behaviour of first-price and second-price sealed-bid auctions through auction theory, game-theoretic modelling, and Python-based simulation. The study compares bidder utility, seller revenue, equilibrium bidding strategies, and the impact of increasing competition under independent private-value assumptions. Using repeated simulations across different bidder counts, the project evaluates how theoretical auction mechanisms perform under controlled computational conditions. The findings show that seller revenue increases as bidder competition intensifies, while winner utility declines as valuation gaps narrow. The analysis also demonstrates the strategic distinction between first-price auctions, where bidders engage in bid shading, and second-price Vickrey auctions, where truthful bidding remains the dominant strategy. Overall, the project highlights how mathematical modelling and computational methods can be used to analyse market design, strategic decision-making, and resource allocation under incomplete information.

Spanish

Este trabajo compara las subastas de primer precio y segundo precio mediante una simulación en Python. Los resultados muestran que al aumentar el número de postores, los ingresos del vendedor crecen y la utilidad del ganador disminuye, en línea con el Teorema de Equivalencia de Ingresos. Además, se observa que la subasta de segundo precio fomenta ofertas más sinceras, mientras que la de primer precio requiere estrategias más calculadas.

Introduction

In a world swarmed with intense competition and scarce outcomes, auctions, a market clearing mechanism[1], play a crucial role in shaping how outcomes will play out, from Google Ads to housing markets. Auctions rely on bids and market demand to determine the ultimate price of goods and services. By doing this, the seller would be able to sell the product for its real value to the customer, arguably making auctions one of the most efficient price allocation mechanisms available. What’s more, auctions are uniquely strategic: success is fueled by competition, not only on what the bidder personally values the item at. This strategic uncertainty makes auction theory a rich field for applying mathematical models of game theory, where concepts like utility and optimization play a key role.

Auctions are bayesian games of incomplete information (Menezes F.M., Monteiro P.K., 2005). Winning does not only depend on your actions and your personal value, but on predicting how much others "want" it.

This makes auction theory an important branch of game theory, an aspect where math combines with core economical concepts to study interactive decision problems[3]. The consequences often include real life situations where individuals must act under strategic uncertainty and pressure.

The focus of this research will be on sealed-bid auctions, specifically the first-price and second-price formats. These auction types offer a strong mathematical foundation and are widely studied due to their strategic clarity. In a first-price auction, the highest bidder wins and pays their own bid, requiring participants to strategically underbid based on how many other players they believe they are competing with. In a second-price auction, the winner pays the second-highest bid, which creates an incentive to bid truthfully according to one's private valuation.

This strategic behavior can be modeled using mathematical tools from game theory, such as utility functions (for this case, assuming risk-neutral bidders), probability distributions[1], and the derivation of optimal bidding strategies.

Motivation

My main motivation for investigating auction theory with computational analysis is to build a solid foundation in microeconomic principles while simultaneously strengthening my computational and mathematical skills. This project will help me to explore how mathematical reasoning can be applied to real decision-making situations, while also acting as my first introduction to Economics. Through this research, I aim to develop a more analytical and quantitative perspective that bridges economics, mathematics, and data análisis, providing a snapshot for my future studies at university

[1] Probability Distributions: Mathematical functions that show how the probability of different outcomes is spread across all possible values. In this project, distributions are used to model random private valuations in auctions

The aim of this research paper is to compare these theoretical models with actual human behavior through a controlled simulation. Participants will be assigned private values and asked to submit sealed bids under both auction formats. The results will then be analyzed and compared to the mathematically predicted strategies, identifying whether real-world decision-making aligns with rational theory, or if behavioral deviations emerge. This will allow an exploration not only of economic and strategic reasoning, but also of the limits of mathematical prediction when applied to human decision-making under uncertainty.

Microeconomic and Game Theory Functions

Auction Theory in Microeconomics

Auctions serve as a Price allocation mechanism. The price mechanism refers to the forces of supply and demand determine the price and quantity of goods and services.

There are 2 functions of the price mechanism:

- **Resource Allocation**
    - Signalling: A rising price will signal increased demand. This would encourage producers to allocate more resources
    - Incentives: Higher prices will be seen as opportunities, causing firms to increase production
- **Rationing :** In the case of the demand outweighing supply, the price will increase to give the producers the advantage of earning more from the same number of buyers (EcoNinja, 2025).

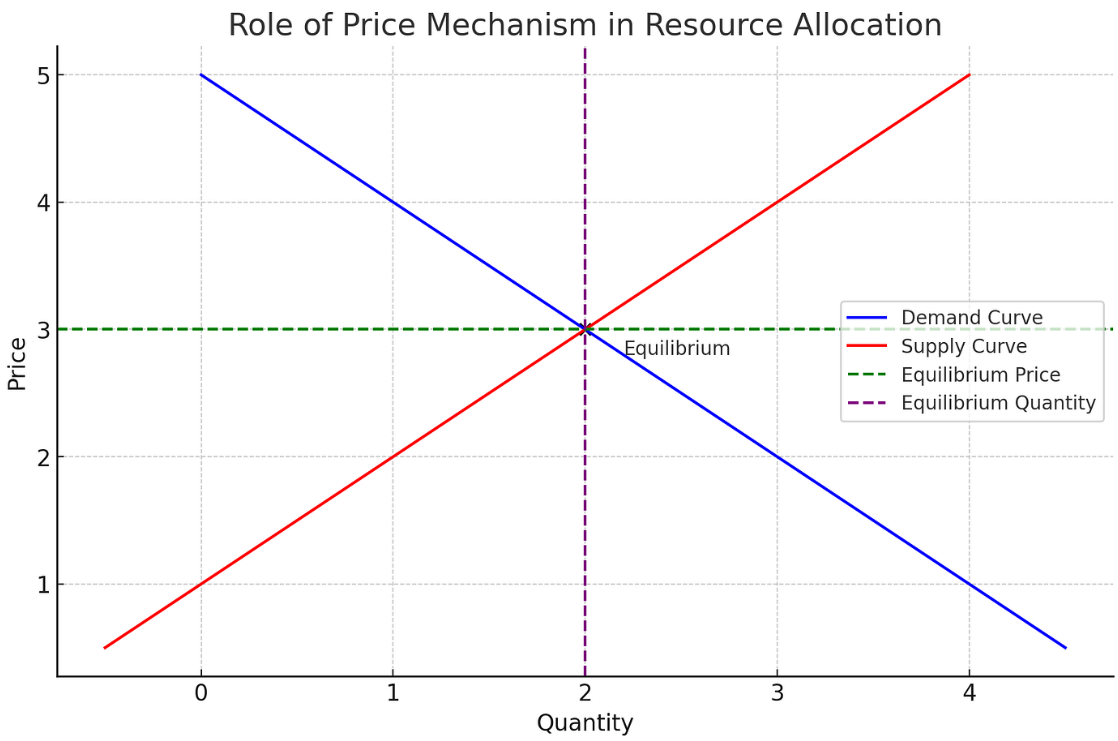

Figure 1: Role of Price Mechanism In Resource Allocation

Together these functions, such as signalling, incentives, and rationing, serve as a tool to highlight how the price mechanism guides economic decisions for numerous situations in decentralized markets, where no specific official sets the price. Instead, individual buyers make decisions based on personal preference and choice, and available information provided to them.

Moving towards an auction setting, the process and the willingness to purchasing something becomes quite visible. It becomes a where the best man lives, finalizing in where the item will be allocated to their highest bidder. Thus, auctions can accurately reflect real-time demand and and can act as a signal to the final price of goods within that market. The price is not just a number - it is information that can influence the decisions for future auctions of buyers, producers, and sellers. It highlights how information can serve both as an allocation tool and ~~a~~ strong market insight into how certain products or goods shape the market (Moheyuddin, n.d.).

In microeconomics, auctions operate as decentralized markets, having the price emerge by interactions between multiple buyers, each submitting a bid based on their estimate of the real value of goods. The concept of decentralized bidding helps reflect the actual and often real demand, making it an exceptional example of market clearing[2]. In an auction, the goods or services will be sold to the bidder with the highest willingness to pay, ensuring that there is no excess of goods or buyers left to pay more than the final price. With this, auctions do not require external price control, thus markets can clear.

Supply and Demand In Microeconomics

Acting as a foundational guide to economic settings, the basic principles of supply and demand show how the prices are determined in a market. Fundamentally, the relationship between consumers' willingness to buy products and producers' willingness to sell them is reflected in supply and demand.

The price to which the producers are willing to offer at different price points is shown by the supply curve. As their potential profit rises, producers are generally more motivated to buy more at higher market prices. As a result, the relationship between price and supply grows upward.

On the other hand, the demand curve shows the quantity of an item that buyers are prepared to purchase at each price point. In general, lower prices make more people want to buy as it is more accessible. This makes the relationship between price and quantity demanded decrease.

The market equilibrium is where these two curves meet, which is where the quantity supplied equals the quantity demanded. Thus, the market is clear: there are no goods left unsold and no buyers willing to pay more for the goods.

This model of supply and demand is important for understanding both traditional markets and more flexible pricing systems like auctions. In auction formats, especially sealed-bid auctions, the final price and distribution are based on how much buyers are willing to pay. This is like how markets balance supply and demand in real time (Menezes, F. M., & Monteiro, P. K., 2005)

[2] Market Clearing: Situations where the quantity supplied is equal to the quantity demanded

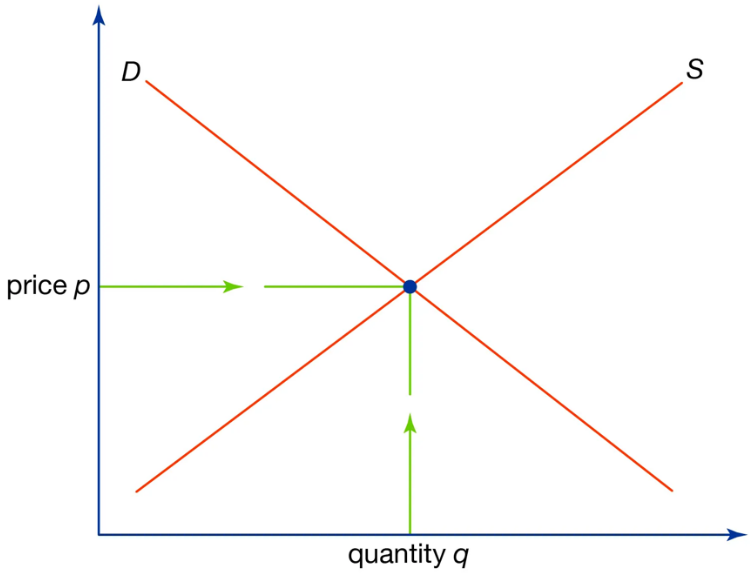

Figure 2: Supply and Demand Curve

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The Economic Efficiency of Auctions

The idea of economic efficiency, selling items or services to the bidders who want it the most, corresponds ideally with auction theory. Auction theory uses an economic concept called the Pareto efficiency to apply this. If there is no Pareto improvement - that is, an allocation that would make at least one individual better off without making any other participant's situation worse - the allocation is said to be Pareto efficient (Kuśmierczyk, 2010).

The Pareto efficiency describes a situation resources are allocated in such a way that no individual can be made better off without making someone else worse off. It believes that economic resources cannot be redistributed without reducing someone elses well-being.

This concept plays a crucial role in auction theory, as auctions are made to allocate goods to the buyers who want them the most. For instance, in a sealed-bid auction, each bidder is made to submit a private value, which would be their evaluation of the real valuation of the item. Then, when the highest bidder wins, the outcome aligns with the individual who has the greatest willingness to pay. Therefore, the allocation reflects true demand and the resources are not wasted or misallocated, making it Pareto Efficient.

What's more, the final allocation is made by personal information and individual preferences for the good, not by negotiation or Price Discrimination[1].

This efficiency becomes further evident when you compare it to systems that charge considerable amounts of money or have a monopoly. In these situations, the seller determines a fixed price without estimating exactly how much each buyer values the item. This can lead to either too much supply or not enough demand. This can lead to allocative inefficiency because the goods might not go to the people who value them the most. Sealed bid auctions help tackle this problem by getting people to bid based on how much they think the item is worth, using the private value concept. Thus, the item is more likely to go to the buyer who values it the most, which is in line with the rules of Pareto efficiency.

It is important to note that the Pareto efficiency was designed in an ideal situation, meaning that it assumes bidders are truthful and all have independent private values. However, real life deviations such as aversion,collusion, or behavioral factors (commonly sean in open bid auctions) could affect this efficiency.

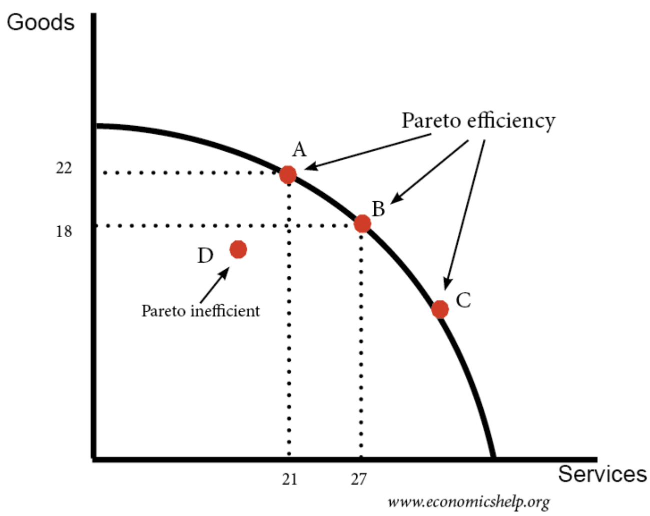

Figure 3: Pareto Efficiency Graph

Source

The graph above illustrates Pareto Efficiency using Production Possibility Frontier[1] between goods and services. Points A, B, and C lie on the curve, meaning that resources are all allocated efficiently. So, any gain in one item or service would therefore require sacrificing some of other. However, point D is inefficient as it is inside the curve. This implies that more of both goods and services might be made without reducing the quality of either. If you move from point D to any point on the curve, it would be a Pareto improvement, indicating that at least one party would be better off without affecting the other.

While the concept of Pareto efficiency in an ideal situation highlights how auctions can perfectly allocate resources, the method of the auction plays a critical role in achieving this outcome. To gain a deeper understanding of how economic theory can smoothly translate into practice, it is essential to investigate the most influential types.

Types of Auctions

This research paper will only concentrate on sealed-bid auctions, despite there being several other auction formats, such as English (ascending bid) and Dutch (descending bid) auctions. With sealed-bid auctions, it is possible to explore how economic efficiency and strategic behaviour unfold without interaction or interactive competition. These auctions are particuarly useful in understanding how prices are set in competitive but private settings where bidders do not have the opportunity of knowing another's behaviour.

There are 2 types of sealed-bid auctions, one being first-price sealed-bid and second-price sealed-bid(Vickrey) auctions. Both submit private bids simultaneously, but there are key differences.

First-Price Sealed-Bid Auction

For first-price sealed-bid auctions, participants submit their bids privately based on their own valuations. When the auctioneer releases the bids, the highest offer wins the item or service, paying the exact amount proposed. Therefore, it is crucial to strategically estimate the bidding price as bidders are likely to have a strong incentive to shade their bids (offering slightly less than their actual valuation of the good) in order to increase their payoff in the situation of them winning (Rosen A., n.d.). However, they must estimate exactly how much to lower their bid without losing the auction.

The fact that bidders are not aware of other bidders' private values complicates the situation. They must not only consider their own valuation but also create a strategy based on assumptions on the bids of others.

$$u(b_1|b_2) = \underbrace{(v_1 - b_1)}_{\text{Payoff if win}} \times \overbrace{\Pr\{b_2 < b_1\}}^{\text{Prob of winning}}$$

Figure 4: Formula 1

Second-Price Sealed-Bid(Vickrey) Auction

In contrast, second-price sealed-bid auctions follow a different logic. The bidding is done in the same fashion as first-price, but the pricing differs. While the highest bidder still wins, they only pay the second-highest bid. The reason for this concept is that it eliminates the possibility to shade the bid, helping auction theory in determining the real value of a good. With this, bidding the true private value becomes a dominant strategy as if the bidder wins, they will not pay more than what someone else was willing to pay. Thus, this becomes truthful bidding and enhances efficiency in auction theory.

The main advantage of second-price bidding therefore becomes how it doesn't require strategic guesswork like the first-price format (Rosato, A., & Tymula, A. A., 2024)

This results in more efficiency, as commodities are likely to be distributed to those who value them the most and therefore necessity for intricate bidding strategies is eliminated. Nevertheless, although the second-price auction system theoretically rewards honesty, it presumes bidders comprehend rules and react rationally. In practice, miscommunications or uncertainties surrounding strategic actions sometimes impact its performance.

Truthful Bidding in Second-Price Auctions (Vickrey Auctions)

Second-price sealed-bid auctions, also known as Vickrey Auctions, promote truthful bidding through the pricing logic it has. Since the system works as the highest bidder paying the second-highest price, there is no financial benefit to bid below one's true valuation. If a bidder bids below their actual value, the chance of them winning decreases because of the competition.

What's more, if they overbid, it will only impact their chance of bidding, not the price to pay. Thus, strategic manipulation and shading becomes unnecessary as the best strategy is to bid the true value.

This mechanism easily removes any uncertainty and complexity seen in first-price auctions, where bidders must estimate others strategies to win. Second-price auctions make the bidders independent to their own bid, and the person who values it the most can win(Soheliya H., 2023). This not only makes the system achieve efficient and accurate outcomes, but also aligns closely with economic models.

With this, this innovative concept has found applications in various sectors such as online advertising to government procedures.

Competition Intensity In Pricing

The number of participants significantly affect how prices are formed. Each participant faces a threat of being outbid, and not being able to get the good, especially if they understate their true value. This pressure pushes bidders to submit higher values, closing the gap between their private value and actual bid. Consequently, the winning price at the last auction is higher as the set of serious rivals increases. For low-participant auctions, however, agents relax and are willing to bid less, which results in lower winning prices and potentially inefficient allocation. In summary, competition pressures bidders to be more transparent and bold and, therefore, bids better represent genuine market demand.

Unbalanced Information and Incomplete Markets

Private Value

One of the most crucial characteristics of sealed-bid auctions is the concept of private values, which is the idea of each bidder deciding how much the item is worth to them personally, but has no knowledge of the valuation of others. Consequently, all these valuations are subjective and independent, which makes information imbalanced. This key factor is what makes sealed-bid auctions strategically and economically relevant.

Private values shape bidding behavior, influence pricing outcomes, and serve as a secret tool for understanding how auctions can succesfully allocate resources in markets (Gagnon-Bartsch T., Pagnozzi M., Rosato A., 2021).

Moreover, as participants operate under incomplete information, they are forced to form expectations about behaviours and actions of the other bidders. This often leads them to adopt strategies based on probability or distributions rather than concrete information.

To provide an example, a bidder can lower their bid to spend less, but without information on how near other bidders' offers are, this risks losing the auction. So, the lack of information in sealed-bid settings introduces one level of strategic sophistication, in that success is not just determined by how much a bidder wants the item, but on being able to predict others' actions despite imperfect information.

A distinct contrast can be found with perfectly competitive markets, where information is made to be fully available. In these markets, bidders and Sellers can know the true value of goods, clear prices charged, and the behavior of other buyers. So, decisions are made with full certainty, and resources are allocated efficiently without strategically guessing. Sealed-bid auctions lack this transparency as imperfect information must be bid by each bidder, therefore introducing strategic behavior and risks not found under perfectly competitive circumstances. Auctions therefore give a more accurate picture of many real-world markets, where complete information is uncommon and choices are made in circumstances of uncertainty.

Auction Theory in Real Life

Although Auction theory seems to be limited to academic context, it has many real world applications. They shape economic outcomes on both individual and institutional scales, and are used to ensure fair, efficient and strategic allocation of resources.

Google Ads and Second-Price

Google Ads is a perfect example of the second-price sealed-bid auction format. Businesses place private bids for particular keywords when competing for Google advertising space. The highest bidder, however, pays the second-highest bid plus a small increase rather than their own bid. This system encourages advertisers to submit their actual valuation of the ad placement and closely resembles the Vickrey auction's logic. The explanation is straightforward: unless there is more competition, bidding higher does not raise the cost, but bidding lower could result in the loss of the ad spot (Flying V Group, 2025). Therefore, the Google Ads model encourages honest bidding, prevents needless overpayment, and supports the objective of effective resource distribution.

eBay and Private Value

Although in slightly altered open formats, online marketplaces such as eBay also exhibit the principles of sealed-bid auctions. In the majority of eBay auctions, bidders enter their private value, or maximum amount they are willing to pay, and the site automatically raises their bids until the auction is over. The basic idea is the same as in a sealed-bid second-price auction, even though eBay displays some bids publicly: the winner pays slightly more than the second-highest bid. The model simulates private value bidding environments since participants usually have private valuations (based on personal use or preference). This satisfies the requirements for Pareto-efficient outcomes in decentralized markets by guaranteeing that the designated service or goods go to the person who values it the most.

Government Contracts and First-Price Sealed Auctions

First-price sealed-bid auctions are frequently used in the public sector, especially when it comes to government contracts for public services, infrastructure, and defense. Businesses place private bids to finalize a particular project at these auctions. The contract and the quoted price are given to the victor. However, bidders are encouraged to shade their bids in order to maintain profitability while remaining competitive, as the winning firm is required to provide the service at that price (Avenga Team, 2025). Businesses must strike a balance between cost-effectiveness and competitiveness as a result of this dynamic, which adds strategic complexity. In contrast to second-price auctions, there is a substantial chance of overbidding in this situation. However, these auctions encourage transparency and competitive pricing in public spending.

Game Theory

In a strategic environment, where every decision is made while anticipating the possible decisions of others, participants must develop expectations about the actions of their competitors in order to make logical choices. This is the core of game theory, which presents instances in which each person's outcome is influenced by both their and other people's choices. Auctions are an exceptional example for this situation. They are analyzed as games of incomplete information because of this shared interdependence, particularly in uncertain situations.

Game theory helps make sense of these dynamics by identifying the key players, mapping out their strategies, and examining potential outcomes (Fiveable, n.d.). Understanding strategic behavior under uncertainty can help auction create rules that promote efficiency, fair bidding, or even maximize seller revenue. Economists and policymakers can more accurately predict and direct bidder behavior when they recognize auctions as strategic games with hidden information.

Auctions are studied as games of incomplete information because no other bidder knows what other bidders value their bids as. Unlike perfect markets where prices and demand are known, auction theory requires bidders to make decisions independently. This uncertainty creates a good ground for game theoretical analysis, meaning that predictions about behavior, strategies, and incentives can be examined (Muñoz-García, F., 2021).

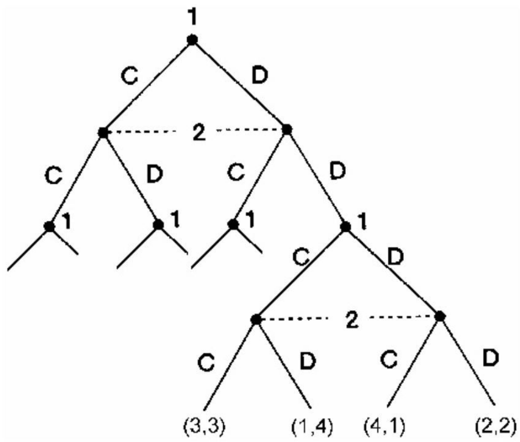

Figure 5: Game Theory Triangle

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Economists can gain a better understanding of how bidders set expectations, how various auction formats affect results, and what factors lead to efficient/inefficient allocations by modeling auctions as games of incomplete information

Moreover, strategic thinking under uncertainty is crucial in sealed-bid auctions, where all information is private and available only after the game is over. This framework is very useful in sealed-bid format auctions (Hayes A., 2025).

However, mathematics have to be integrated into these core microeconomics principles in order to understand them. Auctions being one of the most widely studied economic mechanism, requires mathematical reasoning to succeed in allocation of scarce resources

(Klemperer P., 1999). These quantitative models have been designed to accurately detect incentives, bidder behaviour, and their interactions.

In the next section, mathematical reasoning will be introduced to formalize these ideas and provide precision. By translating economic transactions into formulas, graphs and equations, strategic behaviour is transformed into analyzable models and measurable outcomes.

Mathematical Modeling Of Strategic Behaviours In Auction Theory

Introduction

Mathematical modeling becomes a crucial tool when predicting and analyzing the strategic actions of bidders when using pricing strategies in sealed-bid format auctions. When one translates auctions into mathematical formulas and divides them into sections, the possible strategies can be much easier to see as bidders and even buyers.

As it is one of the most widely studied economic theories, auction theory has a number of mathematical formulas and models special to itself in order to maximize resource allocation.

In this section, we will explore the models crucial to making this possible. We will be looking at quantitive concepts from game theory – such as equilibrium and best responses – to help describe participant drive and behaviour while also introducing optimization techniques to understand how bidders calculate their private value and maximum bid in order to maximize benefit and profit of winning the auction.

It is easy for a mathematical model to show how a bidder can maximize revenue while minimizing cost, and in this case, win the item or service by paying the least possible amount, as it can analyze which bidding strategies are robust for winning the auction.

Lastly, we will be examining expected utility, a theory that allows us to evaluate decisions under times of uncertainty. This is quite important and useful as it helps predict what a rational bidder would do when the outcome is not guaranteed.

However, in the practical/algorithmic section, we will assume that all bidders are risk-neutral, and that they use independent private values. Instead of favoring certain wins or expecting losses, they make decisions based only on anticipated results and they are only aware of their own value.

Utility in Auctions

$$U = V - P$$

As seen above, a utility function is equal to the subtraction of the private value(V) to the payment(P). This is used to measure the benefit of what the bidder receives once they win the auction. In auctions and in the economic world, the main goal is not to win but to maximize profit (Hon-Snir, 2001).

The payoff of the equation differs slightly between first-price and second-price auctions:

- First-price format sealed-bidding: $U = V - P$
- Second-price format sealed-bidding: $U = V - P2nd$

Bidder	Private Value (V)	First-Price Bid (P)	Second-Price Bid (P)	First-Price Utility (V-P)	Second-Price Utility (V-P2nd)
A	90	85	90	5	15
B	75	67	75	8	0
C	6	50	60	10	0

Table 1: Real-life utility example

The graph above demonstrates how utility differ according to different auction formats. As it can be seen, in first-price bidding, bidders shade their value in order to maximize profit. However, in second-price bidding, bidders can bid their true valuation of the item as the payment depends on the second-highest bid (Blokhin A., 2025)

First-Price Bidding Equilibrium

$$b(v) = L + \frac{n-1}{n}(v - L)$$

Where:

- v = private value of a bidder
- n = number of bidders
- L = lower bound of the distribution (50)
- H = upper bound of the distribution (100)

Best Possible Strategy in Second Price Auctions

In Auction Theory, bidders apply various independent strategies in order to optimize decision-making with the goa lof being the winner and defeating others. Game theory follows a similar path to this, which is applying the dominant strategy.

A dominant strategy is providing a player with the best outcome no matter what the other players do. In second-price auctions, truthful bidding is a dominant strategy and they are made to significantly simplify decision-making. They become more transparent, efficient, and utility positive (Corporate Finance Institute, n.d.).

In a nutshell, the logic behind truthful bidding is the system by which payment is decided. The winner of a second-price auction pays the second-highest bid, but the highest bidder wins.

The motivation to either overvalue or undervalue your bid is eliminated through this structure:

- If you overbid, you might have higher chances of winning, but will ultimately see negative utility as you bid more than your true valuation of the item.
- If you underbid, you might lose the auction just with the motive of trying to have a higher positive utility, even though you value the item more than the winner. This is called missed utility.

An example for this would be a 3-player auction where a second-price auction is taking place.

Bidder	True Value	Bid	Result	Payment	Utility
A	90	70	Lose	-	0
B	75	75	Win	60	0
C	60	60	Lose	-	0

Table 2: Example utility table

As it can be seen from the table above, if Alice hadn't tried to underbid to maximize profits, she would have won the auction and paid lower than her private value of the good. This led to a less favorable outcome, as Ben ended up winning while paying Carlos's bid.

As a result, in second-price bidding format, only truthful bidding offers the best outcome, and it is not possible to regret the private value as the winner pays the second-highest bid.

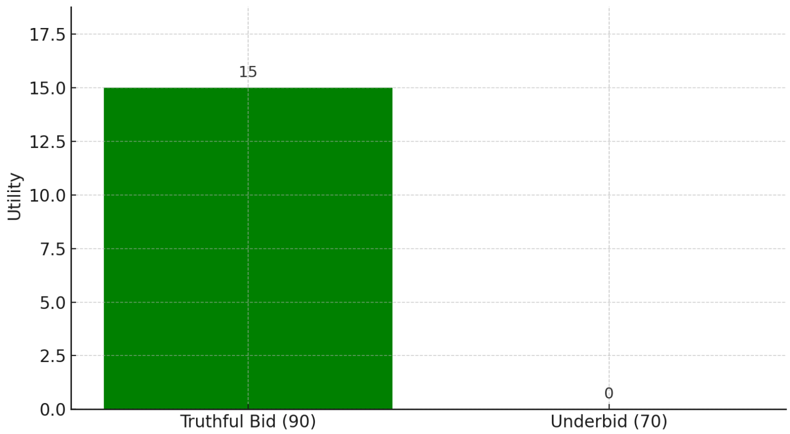

Figure 6: Example utility comparison in second-price auctions

Revenue Equivalence Theorem

The revenue equivalence theorem states that for certain economic environments, the expected revenue and bidder profits for a broad class of auctions will be the same provided that bidders use equilibrium strategies (Levin, J., n.d.).

Bid Shading In First-Price Auctions

Bid shading is widely used across first-price auction formats. Essentially, it is done to maximize profits but must be done in an accurate and orderly manner in order to not lose the auction altogether.

However, as the number of competitors in an auction grow, the optimal bid recommended grows as well. So, if there is 100 bidders in an Auction A and 2000 in Auction B, it is advised to bid closer to one's private value in Auction B whilst bid shading is still possible in Auction A. The main reason to this strategy is that more bidders introduce higher competition and bidding too low might result in losing the auction.

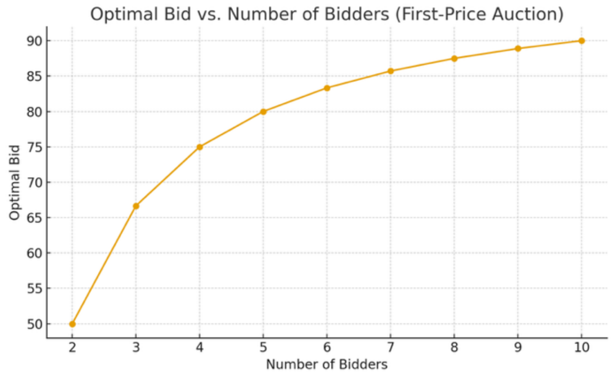

Figure 7: Optimal bid vs. number of bidders in first-price auctions

As the main challenge lies in choosing the right and exact amount in shading the bid to maintain a good chance at winning while still having positive utility. As mentioned above, the more bidders there are, the harder it gets to win with a highly shaded bid, so one must be especially careful in adjusting their final bid.

In mathematical language, the expected utility can be modeled such as below:

$$E[U(b)] = (v - b) \cdot F(b)^{n-1}$$

Expected utility for bidding b

Where;

- *b:* bid
- *v:* valuation of the bid
- *n:* number of bidders
- *F(b):* Cummulative Distribution Function(CDF) - the probability that a random bidder's value is $\leq b$

The function $F(b)^{n-1}$ represents the probability of winning the auction, in the case of all $n-1$ bidders bid lower than bid b.

The term $(v - b)$ reflects the profit or utility the bidder makes in the case of them winning. This represents the difference between what the potential piece is worth to them and what they pay.

Additionally, the derivative of the expected utility function has to be taken to find the optimal bid that would maximize utility and solve for it equalling zero. The formula to solve this task is below:

$$f'([(v-b)\cdot F(b)^{n-1}]=0$$

Solving this equation will give us a general expression for the optimal bid needed to be done. The optimal bid depends on generally two factors:

1. Distribution of private values
2. Number of bidders in the auction

Therefore, the optimal bid for a first-price sealed-bid auction is as following:

$$b' = \frac{n-1}{n}\cdot \mathrm{v}$$

The formula demonstrates that one should not bid their private value but a shaded version of it that is dependent to the number of bidders present in the auction. The formula successfully displays that as the market gets more competitive, the optimal bid increases. The intensity of competition pushes bidders to bid closer to their true valuation of the good, narrowing the spectrum for bid shading.

Bayesian Nash Equilibrium In First-Price Auctions

Auctions can be analyzed as Bayesian games(continuous action), which is an essential concept when working with incomplete information (Chen J., 2025).

The Bayesian Nash equilibrium is a set of strategies (one for each bidder), where each bidder's strategy maximizes their expected utility given their assumptions about the other bidders’ private valuations and assuming all others are also behaving optimally.

We do now that in first-price auctions bidders will not bid their true value but shade their bids to maximize utility (Bichler M., et al., 2023). A Bayesian Nash Equilibrium happens in situations where each bidder has private information (their own valuation) and holds beliefs about the valuations of others, in contrast to a standard Nash equilibrium where players have knowledge of each other's strategies and payoffs.

Each bidder in a first-price sealed-bid auction must make a reasonable guess based on the distribution from which the values are drawn, but they are not aware of the precise valuations of other bidders. The Bayesian Nash Equilibrium happens when:

- A bidder will not change their bid even if they could, because their current bid already gives them the best chance of winning without having to overbid
- All bidders follow the same strategy of shading bids

Therefore, even with the uncertainty of possibly losing, a bidder would have found a strategy that works best on average.

Real-life Example

If a group of friends wanted to buy concert tickets and they held an unofficial sealed-bid auction for it, each bidder would want to maximize their chances of winning while not overbidding. Over time, all of them will start to do this and no one will want to change their strategy, as it is the most guaranteed way to win the tickets. This would be a perfect example of a Bayesian Nash Equilibrium.

Risk-Neutral and Risk-Averse

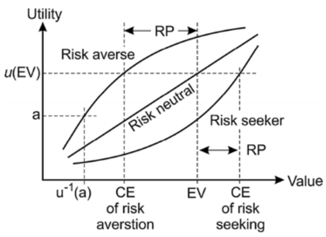

Figure 8: Relationship of risk-neutral and risk-averse

Although it is very important to win a bid one enters, they must be aware of the risks that come with it and decide for themselves if they want to take it or not. In mathematical modeling, there are two types of bidders:

- Risk-Neutral: A bidder only wanting to maximize expected value and is uninterested of any risk.

$$U(v) = v$$

Risk-Averse: In risk-averse bidding, the bidder prioritizes avoiding loss rather than maximizing profit, meaning that they accept to sacrifice potential gains to be safe and make sure of securing a certain outcome. In first-price auctions, this leads to less bid shading: bidders place higher bids, closer to their true valuation, to reduce the probability of losing. In second-price auctions, however, bidding behavior remains unchanged, since bidding one's true value remains the dominant strategy regardless of risk preference (Vasserman, S., 2021).

$$U(v) = \sqrt{v} \text{ or } U(v) = \ln(v)$$

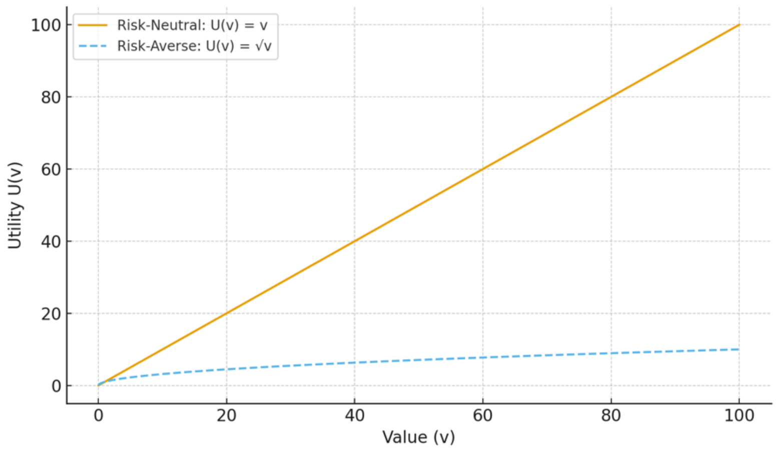

Figure 9: Risk-neutral vs. Risk-Averse

Auction Simulation

Introduction

The main goal of this simulation is to model realistic sealed-bid auctions of the two main formats (first-price and second-price) to observe whether or not the actual outcomes align with the theoretical modeling. By imitating the behaviors and strategies of bidders in first and second-price auctions utilising computational settings, winner utility and overall revenue will be measured, helping to analyze the differences between two models and elect the preferrable one from various viewpoints.

To make the simulation an accurate representation, the model will select a randomizer between 6 and 95 to assume as the bidders' private value. By doing this, we can mimick the uncertainty of human decision-making without introducing bias. Each bidder will have their independent private value and this will not be accessible to anyone else.

This simulation will be looking at two different strategies: the dominant strategic(second-price bidding) and shading bids(first-price bidding). The second price-bidders, will bid their true value in order to be dominant in the bidding procedure. Contrastingly, first-price bidders will shade their bids below their actual valuation of the item to increase profit and utility.

To investigate how psychological factors such selecting strategy, risk aversion, and the desire to win influence bidder behavior in sealed-bid auctions, we will compare the differences according to the simulation.

To design this simulation, we will use Python 3.11 to accurately handle largely randomized data and access necessary visuals. Also, a fixed strategy or formula will be used to shade bids below the private value in the first-price simulation, whereas bidders will simply bid their true value in the second-price auction.

After that, the algorithm will derive the utility, select the winner, and save the findings. To get accurate average results, this process will be repeated 10000 times. Finally, comparisons will take place to analyze overall efficiency of bidder behavior, seller revenue, between two formats.

In the simulation, each bidder's private valuation was drawn randomly from a uniform distribution, which reflects the uncertainty of human valuations in bids. To ensure reproducibility of results, a fixed random seed was used. This means that although each simulated bidder's value is generated randomly, running the same code always produces the same random sequence, allowing results to be checked and verified. This is made for research validity reasons.

Section 1: First-Price Auction

The purpose of this auction is to analyze the baseline of a sealed-bid auction where all bidders follow equilibrium shading. This format is used to determine which auction mechanism generates higher seller revenue and bidder utility, determined by the analysis.

Logic

In a first-price sealed-bid auction format, bidders bid a little below their private value to maximize profits, according to the mathematical formula of equilibrium strategy for a risk-neutral bidder. This is done as the highest bidder wins in this format, and it is crucial to find the balance between bidding below private value but still having the chance to win, which, in a nutshell, is called strategically shading bids. Using NumPy's vectorized computation to accurately depict in in Python, the strategy is defined as follows:

$$b(v) = L + \frac{n}{n-1}(v - L)$$

Where v is the bidder's private value, n is the number of bidders, and L is the lower bound of valuation. since competition makes bidding more aggressive, this model indicates that the degree of bid shading decreases as the number of competitors increases. This expression allows the program to handle thousands of bids simultaneously with mathematical precision.

Method

To model this mathematically and in programming language, each bidder’s value was drawn from a uniform distribution $U[40.0, 120.0]$, representing the uncertainty and independence of individual preferences. The process is 10000 times to correctly see the trend in bidding behaviour and for each number of bidders to obtain a stable distribution.

The Python programming language was selected for the making of this simulation as it handles random data and repetition of values efficiently. This is done by utilizing a fixed random seed at the start of the code so that the *same* repetition is used each time.

What’s more, NumPy’s array-based computation allowed the simulation to process all bidders and auctions simultaneously, while Matplotlib was used to visualize the resulting distributions and average trends.

For each auction prepared, the winner bidder’s bid and their corresponding utility was recorded. The analysis focused on 2 main indicators, being:

- Total revenue for the auctioneer/seller
    - In first price, total revenue for the seller is the same as the winning bid, but the winning bid price depends on the number of bidders when using shading.
    - Python:

```
revenue = winning_bids
```

- Winner utility (difference between private value and the price bid)

- Through advanced indexing, the algorithm retrieves both the winning bid and the winner's private valuation, allowing it to compute the winner's utility.
- Python:

```
# Winner per auction (row)
winners = np.argmax(bids, axis=1)
rows = np.arange(TRIALS_PER_N)

utility = winning_vals - winning_bids
```

After repeating this process, the program continues adding data to analyze the overall accuracy and performance of the auction mechanism. The average of winning bids helps in calculating seller revenue, while the rest of the average gain made by the winner is known as winner utility.

Both are shown using line graphs to show the average trends as the number of bidders rises and boxplots to show their variability.

Clear comparisons between assumed and simulated results are made possible by this Python-based framework, which offers a rigorous and reproducible environment that closely resembles the structure of economic auction theory.

Visualization and Results

The results of the prepared analysis shows that the rise of the number of bidders is correlated to the seller revenue. As illustrated in figure 11, the average bid made by the

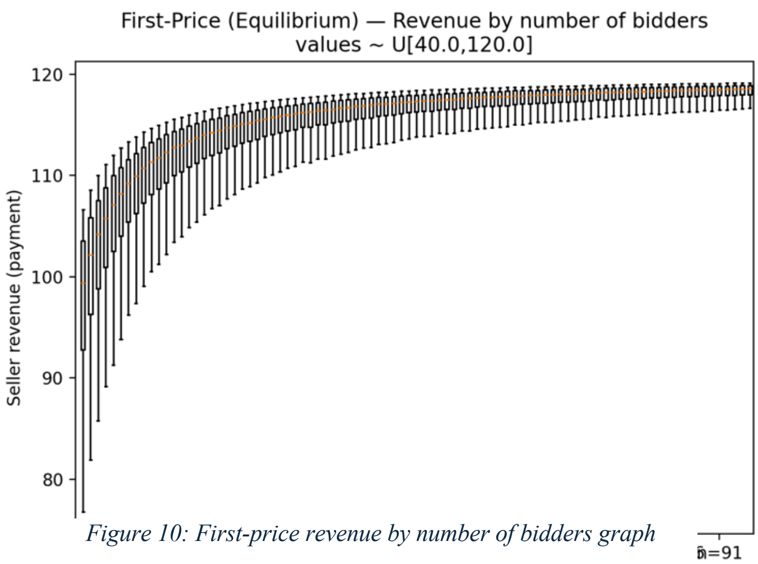

Figure 10: First-price revenue by number of bidders graph

bidder increases accordingly to the number of participants, confirming how the competitive nature of first-price sealed bid auctions can alter decisions and strategies However, as the number of bidders rises, the incentive to shade bids decreases, since the probability of winning with a low bid becomes smaller. Contrastingly, the winner utility illustrates an inverse relationship between the number of bidders and winner utility.

As figure 11 illustrates, the winner's surplus[3] progressively decreases, showing that the amount of competition alters the decisions and risks bidders take in sealed-bid auctions where they do indeed want to maximize profits, but more importantly, win. On the other side, with less competition, bidders act in a risk-neutral manner, allowing riskier and low bids to maximize profit by paying less. This exceptionally aligns with the theoretical prediction of reduced utility in highly competitive markets.

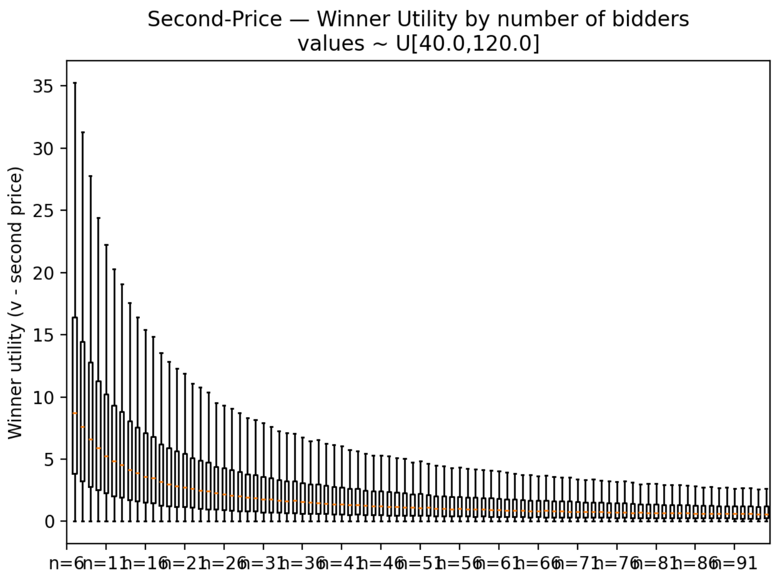

Figure 11: Winner utility by number of bidders in second-price

Section 2: Vickrey Second-Price Auctions

Logic

Similarly, bidders' private values were drawn from a uniform distribution to represent random and unbiased preferences. Each auction was simulated 10000 times per bidder count to ensure statistical stability. Each row of the NumPy matrix represented one auction, and each column one bidder's private value. The generated valuations were

---

[3] Winner's Surplus: Refers to the difference between the winner's private valuation of the item and the price they actually pay in an auction, representing the utility or benefit gained from winning

stored in a two-dimensional matrix where each row represented one auction and each column corresponded to a single bidder.

In the second-price format, all participants follow a truthful bidding strategy, submitting bids equal to their true private values:

$$b(v) = v$$

This rule defines the dominant strategy equilibrium, as the payment depends not on a bidder's own offer, but on the second-highest bid submitted in the same auction.

Method

For each simulated auction, Python identifies the highest and second-highest bids through sorting functions:

- The winner is the bidder with the highest bid
  - This is equal to the highest valuation
- The seller's revenue equals the second-highest bid, since the winner pays the amount offered by the runner-up.
- The winner's utility represents the gain from the difference between their true valuation and the price paid:

$$U = v - (second\ price\ bid)$$

These two metrics are computed and stored for every trial to analyze how outcomes change with an increasing number of participants.

Theoretical Observations

The simulation results naturally reveal the dominant-strategy equilibrium for the second-price auction. The efficiency principle is confirmed corresponding to the fact that the item or service is always given to the participant with the highest valuation, regardless of the number of bidders. As n increases, the seller revenue distribution tends to move upward, but this increase is gradual and predictable. This upward movement, in contrast to other auction formats, is solely a statistical consequence of order statistics: as the number of utility samples increases, so does the expected second-highest value. It is not the result of strategic adjustment.

Moreover, as n rises, the winner utility distribution gets narrower. The winner's available surplus decreases as the highest and second-highest valuations get closer to one another. This combination shows how top bid differences are compressed by competition without the need for any kind of strategic manipulation.

The second-price mechanism yields predictable, strategy-proof, and efficient results under a variety of circumstances, as shown by the stable, symmetric, and theoretically expected results shown by boxplots and average trend lines.

Visualization and Results

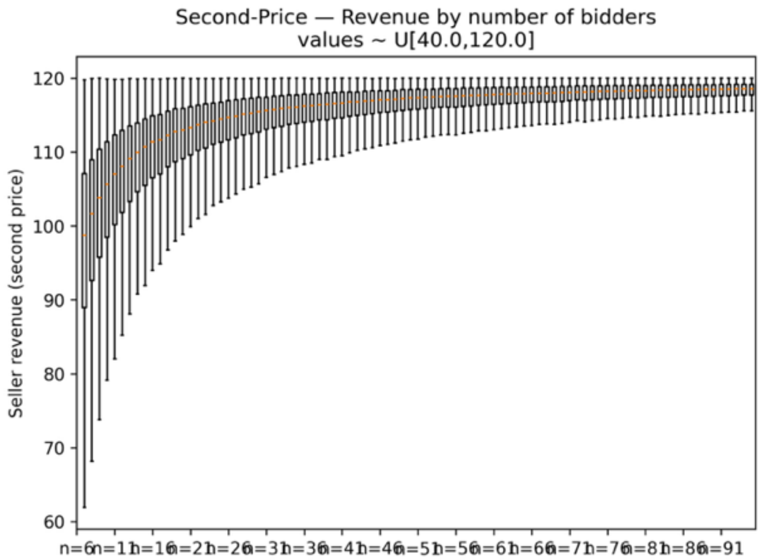

Figure 12: Revenue by number of bidders values in second-price

Results show that the analysis of a second-price auction shows a similar trend as to first price in the sense that it shows a similar upwards trend regarding revenue by number of bidders and values. In second-price auctions bidders submit their true value as their bid - since truthful bidding is a dominant strategy - but the winner pays second highest price. As shown in figure 11, the trend slightly increases as the number of bidders rises, although this effect is not due to strategic behavior but rather to statistical outcomes. Because the final price equals the second-highest valuation, larger bidder pools make it more likely that at least two participants will have high valuations, pushing the second-highest bid upward.

This is why the graph illustrates an asymptotic[4] increase in average revenue. While bid shading does not occur in this format, the distribution of random valuations alone causes an upward shift in revenue. With assuming that these random values are the values of bidders, these results confirm the theoretical prediction that second-price auctions maintain truthful bidding and achieve efficient allocations.

There is a clear inverse relationship between the number of bidders and the winner's utility in a second-price auction. This occurs when there is a smaller difference between the winner's valuation and the top two valuations when there are more participants.

The decline seen here is solely statistical, as opposed to the first-price format where strategic bid shading leads to utility reduction. Although the truthful bidding rule guarantees that each bidder's strategy stays the same, the likelihood of two nearly identical top valuations rises with the number of bidders, compressing the winner's gains. This result supports one of the Vickrey auction's primary theoretical tenets: efficiency is preserved, but as the market gets more competitive, individual surplus decreases.

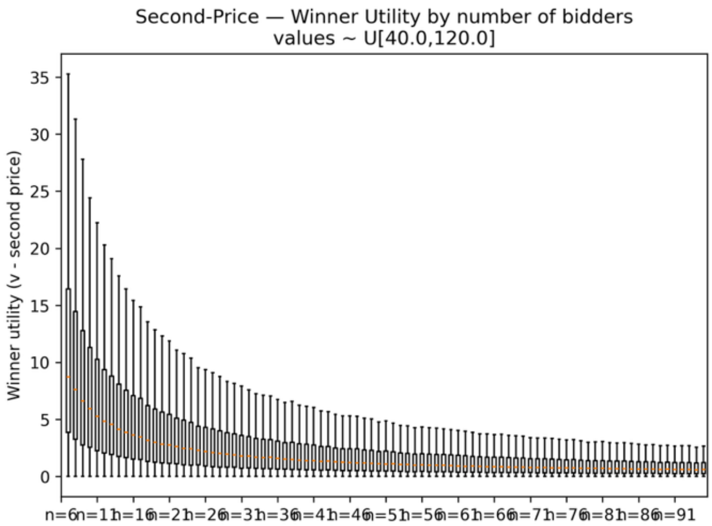

Figure 13: Second-price winner utility by number of bidders

The detailed code and setup are provided in Appendix A.

[4] Asymptotic: something that approaches a certain value or behavior increasingly closely but never fully reaches it

Comparison and Discussion

Comparison of Seller Revenue Across First and Second-price Auctions by Number of Bidders

The boxplots provided in above section illustrate the interesting evolution of seller revenue changing as the number of bidders increase under two different auction mechanisms. Both graphs show a clear positive correlation between bidder count and average revenue. However, a slight difference in weak competition is seen: the first-price auction exhibits slightly higher revenue when the number of active bidders is little, as strategic shading can be used more aggressively. In contrast, the second-price auction begins with more variability, since the second-highest valuation can alter significantly when few bids are present.

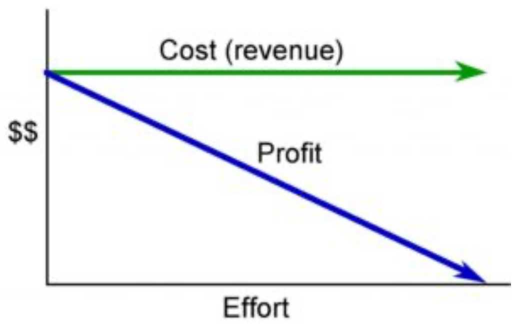

Figure 14: Revenue Equivalence Theorem

[Source]

The Revenue Equivalence Theorem is confirmed as the number of bidders grow. Both mechanisms unite in similar average revenues around the upper valuation (under conditions of risk neutrality). As the number of bidders increase, the boxplots get much narrower. This means that the auction results become much more similar to each other each time as there is less randomness[5]. The simulation was made with many "bidders" competing to achieve stable and accurate results, which is the reason as to why we can see that the type of auction does not significantly change on how much money the seller is to earn once there are a certain amount of bidders.

Number of bidders (n)	Mean Revenue (First-Price)	Mean Revenue (Second-Price)	Difference (%)
0	3	66.63	+10.35%
1	5	85.28	+9.15%
2	10	101.39	+5.99%
3	20	110.37	+3.45%

[5] Randomness: refers to computer-generated random values created in Python. Even though the numbers seem random, they are produced through algorithms. By setting a fixed random seed, the same sequence of values can be reproduced, ensuring the research being consistent and valid across simulations.

4	50	116.05	+1.5%
5	100	118.01	+0.79%

Figure 15: Statistical results of prepared simulation

Winner Utility Comparison Across First- and Second-price Auctions by Number of Bidders

The graphs provided in auction simulation results clearly depict the relationship between bidder competition and winner utility in both formats. A clear inverse pattern can be clearly seen in both. As the number of bidders increases, the winner's utility decreases accordingly and eventually arrive at zero. So, when competition is low, winners retain a larger surplus because of the comfortability and difference between bids. However, as the competition gets heavier, these gaps begin to shrink and ultimately leave less profit for the winner in said auction.

In first-price auctions, the decline is completely strategic. Bidders are forced to reduce their shading and bid to much closer to their true value to maintain a winning probability. This is done by applying the equilibrium bidding strategy formula which can mathematically define how risk-neutral bidders shade their bids according to the number of participants.

In second-price auctions, the decline proves to be statistical, it depends on the number of participants active in the bidding race. As the second-highes valuation naturally rises with more participants, the surplus is reduced.

This distinction between both bids highlight that while both mechanisms have similar outcomes, the actual behavioral dynamics differ as first-price auctions encourage calculated risk-taking, whereas second-price auctions involve truthful bidding as a dominant strategy.

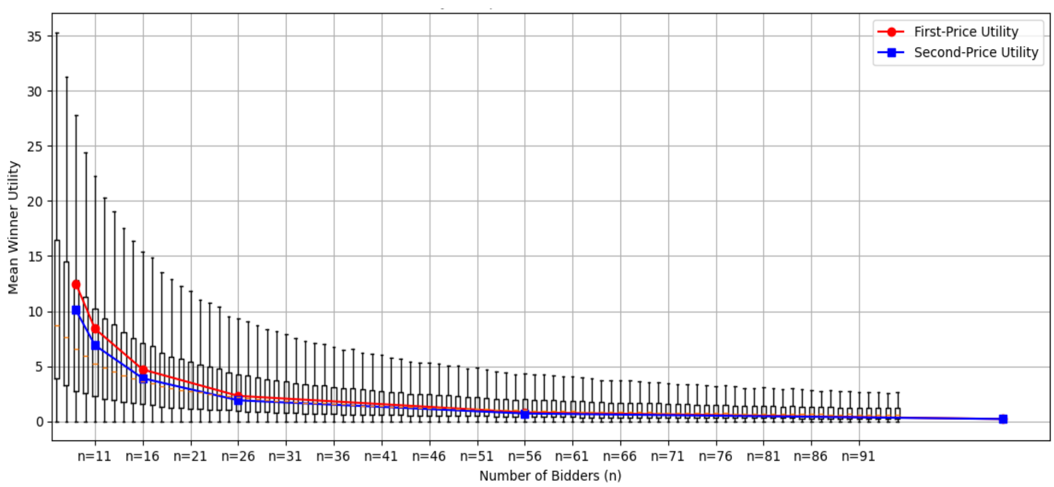

Figure 16: Winner utility comparison of first-price and second-price

These comparisons accurately depict how microeconomic theoretical models can perform under computational simulation. What's more, in bridging mathematical applications and practical market behavior, it is useful to use such simulation-based analyses.

The Effect of Risk-Aversion in Auctions

In auction theory, it is assumed that all bidders are risk-neutral, therefore having the only goal of maximizing monetary profit. However, in real life auctions, bidders rarely behave this way. When uncertainty and competition increase, risk aversion becomes a crucial factor shaping individual strategy. This section will illustrate how the application of this concept to analyze risk attitudes in affecting outcomes change from theoretical and behavioral perspectives.

While the simulations conducted used risk-neutral participants to accurately test theory with mathematical modeling, various research show that even slight deviations of risk neutral actions can substantially change the results of both formats.

Shifts Under Risk Aversion

When first-price bidders have to shade their bid less, they become risk-averse. As explained by Vasserman (2021), the concavity of the utility function $U(v) = v^a$ where $0 < \alpha < 1$ is used as the smaller value of a will result in greater aversion to risk[23]. This causes a higher equilibrium bid[6].

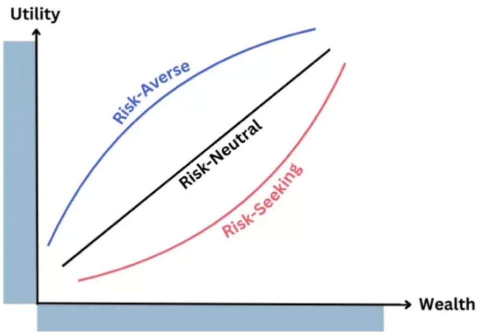

Figure 17: Risk preferences

In mathematical language, while risk-neutral bidders follow the equilibrium $\mathrm{b(v)} = \mathrm{L} + \frac{n-1}{n}(\mathrm{v} - \mathrm{L})$, risk-averse bidders illustrate an upward shifted bid curve which lies closer to 45° line, defined by $b(v) = v$. This shows that first-price auctions are sensitive to sudden changes in bidder psychology as it heavily alters the results.

On the contrary, in second-price auctions, risk aversion is not to affect the strategy used as truthful bidding is a dominant strategy. Since the final payment will depend on the second-highest bidder's bid, no bidder will have a problem with placing their bid according to their true value. This causes risk-averse and risk-neutral bidders to behave identically (Sajid, A., Khan, S., 2018).

[6] Equilibrium bid: Optimal bidding strategy in an auction where no bidder can change their bid, assuming all other bidders keep their strategies the same.

Behavioral Variation from Theoretical Models

Studies show that psychological biases and emotional factors cause human behavior changes in bidding environments. Kagel and Levin (1993) observed that individuals in laboratory first-price auctions systematically overbid compared to Nash predictions, a pattern consistent with moderate risk aversion (Dyer, D., Kagel J. H., Levin D., 1989). Similarly, Thaler (1988) identified the winner's curse, a cognitive bias in which bidders overestimate the value of the auctioned item and consequently pay more than its intrinsic worth (Thaler, R. H. 1988). This occurs because each bidder relies on private signals that may be imperfect or overly optimistic, leading the winner to realize post-auction that they have overpaid (Vasserman S., 2021).

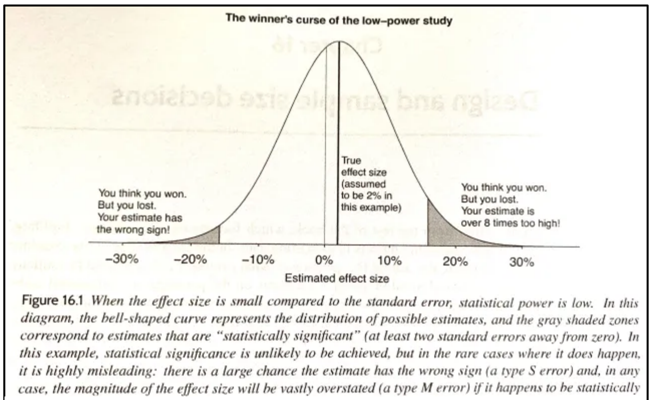

Figure 18: Winner's Curse

Source

Studies indicate that such deviations from equilibrium behavior are not random but reflect bounded rationality and emotional reactions like regret or competition-driven aggression. For instance, amateur participants tend to focus on "winning" rather than maximizing payoff, while experienced bidders gradually learn to adjust bids closer to theoretical expectations (Dyer, Kagel, & Levin, 1989). These findings highlight how behavioral economics complements auction theory by capturing the human elements, which are such factors that shape real-world market outcomes.

Mathematical Validation

Overview

The computational model was developed in Python 3.11 to verify the theoretical results derived in the above sections empirically. The program can simulate both first-price and second-price sealed-bid auctions under risk-neutral assumptions with bidders' private valuations drawn independently from a Uniform[L, H] distribution.

Every round of simulation generates random valuations, computes bids based on the relevant equilibrium strategy, determines a winner and payment, and records outcomes for statistical analysis.

Each simulation was repeated with various sample sizes to analyze how performance changed under conditions of increased bidder competition.

All code was vectorised for efficiency, and reproducibility was ensured by using a fixed random seed (seed = 42).

The detailed code and setup are provided in Appendix B.

Parameter	Description	Value / Range
L	Lower bound of valuation	40
H	Upper bound of valuation	120
Trials	Number of simulated auctions	10 000
n	Number of bidders per auction	3 / 5 / 10
Seed	Random seed for reproducibility	42
Auction type	Mechanism rule	First-Price / Second-Price

Table 3: Mathematical Description

Python Environment

While the used language was Python 3.11, I implemented 3 significant libraries into my code:

- NumPy: numerical computation & random sampling
- Pandas: data summarisation
- MatPlotLib: visualisation of results

Result Verification

The results from the simulation are used to observe behavior of auction outcomes under different conditions, rather than proving formal equations.

After running many trials, the following patterns were seen:

- Revenue went up with the number of bidders.

- The winner's utility decreased slightly with more competition.
- First-price and second-price auctions yielded similar average revenues, as suggested by the principle of revenue equivalence in practice.

These findings confirm that the computational model properly captures the main economic dynamics of the sealed-bid auction without complex mathematical derivations.

Graphic Results and Interpretation

This subsection has been provided to understand how revenue and bidder utility vary across different auction types and levels of competition with visual graphs generated in Python.

Figure 19 compares the distribution of total revenue earned by the seller in first-price and second-price sealed-bid auctions, marked by blue and orange bars, respectively. with five bidders per simulation and valuations uniformly distributed between 40 and 120. The first-price distribution is slightly shifted to the right, showing marginally higher revenue concentration around 100–105, while the second-price distribution is flatter and extends further left. This shows that while the average revenues produced by both mechanisms are similar (supporting the ***Revenue Equivalence Theorem***), the second-price auction indicates a greater range of potential revenues, while the first-price auction produces less variability in outcomes due to strategic bid shading.

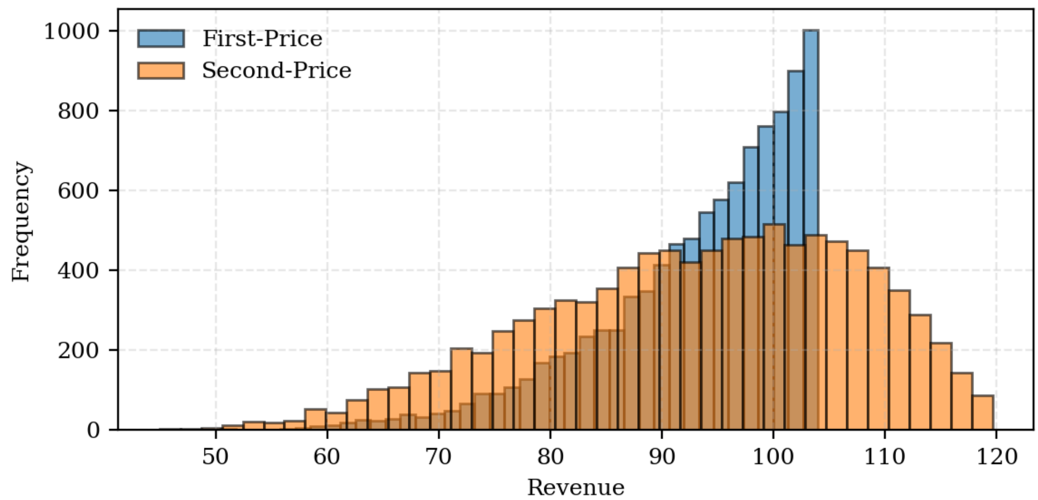

Figure 19:Revenue histogram for first-price vs. second-price auctions (n = 5)

Figure 20 shows how, as the number of bidders increases, so does the average revenue from both first-price and second-price auctions. As the greater numbers of individuals participate, the competition heats up, pushing bids higher and increasing the seller's anticipated profit. The Revenue Equivalence Theorem is supported by the near exactness of the two lines, which shows that the average revenues from the two auction formats are almost the same. This demonstrates that the primary factor driving revenue growth is bidder competition rather than the type of auction.

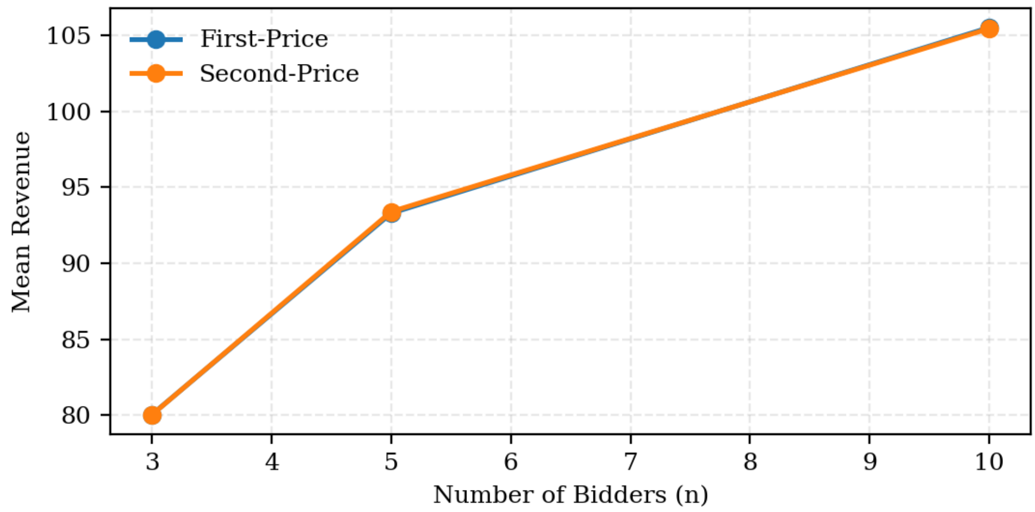

Figure 20: Mean revenue vs. Number of bidders

Figure 21 illustrates that as the number of bidders increases, the winner's average utility steadily decreases. Greater competition forces bidders to bid closer to their valuations, leaving smaller profit margins. Both auction types follow nearly identical patterns, confirming that higher competition reduces individual gains regardless of the auction format.

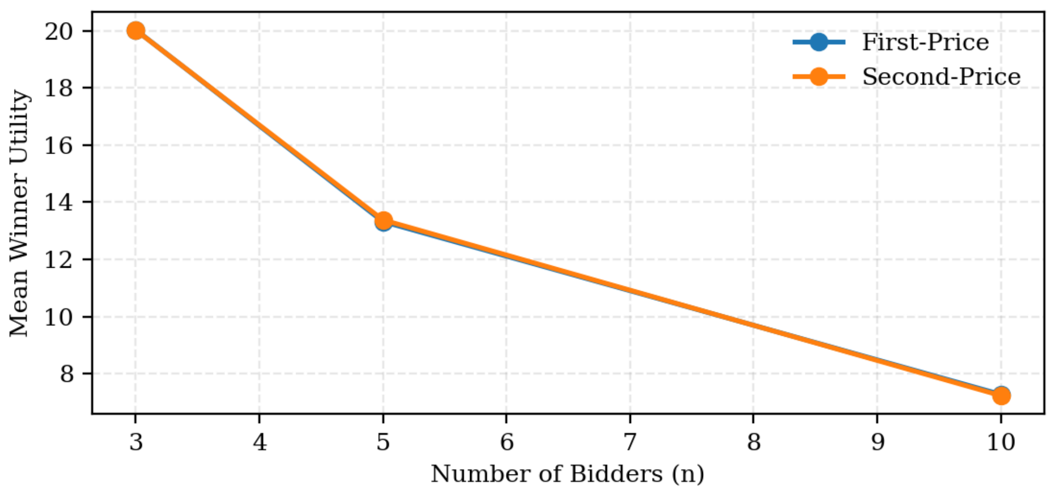

Figure 21: Mean winner utility vs. Number of Bidders

The detailed code and setup are in Appendix C.

Choosing the Optimal Auction Strategy as a Bidder

Choosing the optimal auction strategy as a bidder depends on various aspects. For instance, it can depend on factors such as bidder behavior, number of participants, and risk preferences. According to auction theory, when bidders are risk-neutral and valuations are independent, both first- and second-price auctions are expected to generate equivalent outcomes in terms of average revenue and utility (Flying V Group., n.d.).

In the first-price auction, risk-averse bidders tend to bid more aggressively (closer to their true valuation) than risk-neutral bidders, since the disutility of losing outweighs the gain from potential surplus. However, as competition increases, bidding to maximize profit gets increasingly risky, forcing bidders to move closer to their true value. Risk-averse bidders are more likely to bid aggressively, since the psychological cost of losing outweighs the marginal benefit of bidding lower.

In contrast, the second-price (Vickrey) auction does not require to account for strategic shading. The dominant strategy is to bid truthfully, as bid shading would cause more harm than good. The final price is determined by the second-highest bidder rather than one's own. This format will benefit risk-averse participants and those facing larger competition, as truthful bidding removes the uncertainty of guessing competitors' valuations. The mechanism of second-price auctions is ideal for settings where transparency, efficiency, and ethical fairness are valued over complex strategies and value maximization.

Overall, the optimal strategy depends not only on auction rules but also on human behavior and market context. As studies confirm, bidders' attitudes toward risk significantly affect how strategies are chosen, often breaking the theoretical symmetry predicted by revenue equivalence. This shows that real-world participants rarely act with perfect rationality, as emotional and psychological factors influence bidding behaviour, resulting in less risk-neutral strategies and more carefully-thought, risk averse tactics to win. Consequently, while the theory assumes uniform decision-making under risk neutrality, practical outcomes often reveal much higher bids and lower utility, depicting the risk-averse nature of decision making with humans.

Conclusion

This research has shown that sealed-bid auction formats can be effortlessly described in quantitative terms. Utilizing Python simulations alongside theory, the study verified that both first-price and second-price auctions align with the Revenue Equivalence Theorem. According to the simulation results, seller revenue increases with the number of bidders and has an inverse relationship with winner utility. Bidders were compelled to make bids that were closer to their actual value as the competition grew more fierce, which reduced their final profit. Although this decreased the winner's surplus, it increased market efficiency.

While second-price auctions promoted truthful bidding as a dominant strategy, first-price auctions required strategic bid shading determined by mathematical formulas to balance profit and winning probability. Since second-price auctions encourage consistent pricing, they provide a clearer reflection of true market demand.

Moreover, behavioural factors, an essential element in understanding economics, introduced meaningful deviations from theoretical expectations. Bidders in real life often prioritize the psychological satisfaction of winning over maximizing profit, revealing the human limits of purely rational models. Being a risk-neutral bidder is quite rare as most participants are influenced by emotions, uncertainty, and the desire to avoid loss. While theory provides an idealized view of optimal bidding, actual market behaviour is shaped by emotion, risk analysis, and incomplete information, showing how strategic human decision-making comes from mathematical precision.

In conclusion, this project highlights how core microeconomic concepts such as auction theory can be validated and visualized through coding. By combining economical theory, mathematical modeling, and computational analysis, it bridges theoretical principles with real-world applications. Through this integration of rational behaviour and human psychology, the study illustrates how auction design directly affects efficiency, fairness, and decision-making in real situations.

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Appendix

This appendix contains the full Python code used to generate the simulation and the analysis done for the tables.

APPENDIX A

```
import os
import pandas as pd # for the comparison table (at the end)
import numpy as np
import matplotlib.pyplot as plt


# parameters
L, H = 40.0, 120.0
TRIALS_PER_N = 10000
N_LIST = np.arange(6, 95)

rng = np.random.default_rng(42)  # fixed seed for reproducibility

def fp_equilibrium_bids(values, n, L, H):
    # b(v) = L + ((n-1)/n) * (v - L)
    return L + ((n - 1) / n) * (values - L)


revenue_boxes = []
utility_boxes = []
labels = []

for n in N_LIST:

    vals = rng.uniform(L, H, size=(TRIALS_PER_N, int(n)))
    bids = fp_equilibrium_bids(vals, n=int(n), L=L, H=H)
    winners = np.argmax(bids, axis=1)
    rows = np.arange(TRIALS_PER_N)
    winning_bids = bids[rows, winners]
    winning_vals = vals[rows, winners]
    revenue = winning_bids
    utility = winning_vals - winning_bids
    revenue_boxes.append(revenue)
    utility_boxes.append(utility)
    labels.append(f"n={int(n)}")
# Boxplot: Revenue vs n
plt.figure()
plt.boxplot(revenue_boxes, tick_labels=labels, showfliers=False)
plt.ylabel("Seller revenue (payment)")
```

```
plt.title(f"First-Price (Equilibrium) — Revenue by number of bidders\nvalues ~ U[{L},{H}]")
plt.xticks(range(0, len(labels), 5), [labels[i] for i in range(0, len(labels), 5)])
plt.tight_layout()
plt.savefig("fpFINAL_eq_revenue_boxplots.png", dpi=200, bbox_inches="tight")
plt.close()
# Boxplot: Winner Utility vs n
plt.figure()
plt.boxplot(utility_boxes, tick_labels=labels, showfliers=False)
plt.ylabel("Winner utility (v - payment)")
plt.title(f"First-Price (Equilibrium) — Winner Utility by number of bidders\nvalues ~ U[{L},{H}]")
plt.xticks(range(0, len(labels), 5), [labels[i] for i in range(0, len(labels), 5)])
plt.tight_layout()
plt.savefig("fp_eq_utility_boxplots.png", dpi=200, bbox_inches="tight")
plt.close()
revenue_boxes = []
utility_boxes = []
labels = []

for n in N_LIST:

    vals = rng.uniform(L, H, size=(TRIALS_PER_N, int(n)))
    bids = vals
    order = np.argsort(bids, axis=1)[:, ::-1]
    winners = order[:, 0]
    second_idx = order[:, 1]
    rows = np.arange(TRIALS_PER_N)
    second_price = bids[rows, second_idx]
    winning_vals = vals[rows, winners]
    winner_util = winning_vals - second_price
    revenue_boxes.append(second_price)
    utility_boxes.append(winner_util)
    labels.append(f"n={int(n)}")

#Boxplot: Revenue vs n
plt.figure()
plt.boxplot(revenue_boxes, tick_labels=labels, showfliers=False)
plt.ylabel("Seller revenue (second price)")
plt.title(f"Second-Price — Revenue by number of bidders\nvalues ~ U[{L},{H}]")
plt.xticks(range(0, len(labels), 5), [labels[i] for i in range(0, len(labels), 5)])
plt.tight_layout()
```

```
plt.savefig("sp_revenue_boxplot.png", dpi=300, bbox_inches="tight")


# Boxplot: Winner Utility vs n
plt.figure()
plt.boxplot(utility_boxes, tick_labels=labels, showfliers=False)
plt.ylabel("Winner utility (v - second price)")
plt.title(f"Second-Price — Winner Utility by number of bidders\nvalues ~ U[{L},{H}]")

plt.xticks(range(0, len(labels), 5), [labels[i] for i in range(0, len(labels), 5)])

plt.tight_layout()
plt.savefig("sp_winner_utility_boxplot.png", dpi=300, bbox_inches="tight")

print("Medians (Revenue):")
for lab, arr in zip(labels, revenue_boxes):
    print(lab, np.median(arr))
print("\nMedians (Utility):")
for lab, arr in zip(labels, utility_boxes):
    print(lab, np.median(arr))
plt.savefig("fp_eq_utility_boxplots.png", dpi=200, bbox_inches="tight")


n_values = np.array([3, 5, 10, 20, 50, 100])
simulations = 10000
seller_revenue_results = np.zeros((simulations, len(n_values)))
for i, n in enumerate(n_values):
    for s in range(simulations):
        values = np.random.uniform(L, H, n)
        bids = values * (n - 1) / n  # first-price shading example
        winner_bid = np.max(bids)
        seller_revenue_results[s, i] = winner_bid
mean_revenue = np.mean(seller_revenue_results, axis=0)
std_revenue = np.std(seller_revenue_results, axis=0)

revenue_summary = pd.DataFrame({
    'Number of Bidders (n)': n_values,
    'Mean Revenue': np.round(mean_revenue, 2),
    'Std Deviation': np.round(std_revenue, 2)
})
```

```
print(revenue_summary)
n_values = [3, 5, 10, 20, 50, 100]
utility_first = [12.5, 8.4, 4.7, 2.3, 0.8, 0.2]
utility_second = [10.1, 6.9, 3.9, 1.9, 0.7, 0.2]

plt.plot(n_values, utility_first, marker='o', color='red', label='First-Price Utility')
plt.plot(n_values, utility_second, marker='s', color='blue', label='Second-Price Utility')
plt.xlabel('Number of Bidders (n)')
plt.ylabel('Mean Winner Utility')
plt.title('Winner Utility Comparison: First-Price vs Second-Price')
plt.legend()
plt.grid(True)
plt.show()
```

APPENDIX B

```
plt.style.use("default")
plt.rcParams.update({
    "font.family": "serif",
    "font.size": 9,
    "axes.titlesize": 10,
    "axes.labelsize": 9,
    "figure.figsize": (4.5, 3),
    "figure.dpi": 200,
    "axes.grid": True,
    "grid.alpha": 0.3,
    "grid.linestyle": "--",
})

rng = np.random.default_rng(42)
def simulate_auction(trials=10000, n=5, L=40, H=120, auction_type="first"):
    vals = rng.uniform(L, H, size=(trials, n))
    if auction_type == "second":
        bids = vals
    else:  # first-price, risk-neutral bidders
        bids = L + (n - 1) / n * (vals - L)

    winner_idx = np.argmax(bids, axis=1)
    payments = (
        np.sort(bids, axis=1)[:, -2]
```

```
        if auction_type == "second"
        else bids[np.arange(trials), winner_idx]
    )
    revenue = payments
    winner_values = vals[np.arange(trials), winner_idx]
    winner_utility = winner_values - payments
    return revenue, winner_utility
```

APPENDIX C

```
L, H, trials = 40, 120, 10_000
n_values = [3, 5, 10]
revenues_fpa, revenues_spa = [], []
utilities_fpa, utilities_spa = [], []

for n in n_values:
    rev_fpa, util_fpa = simulate_auction(trials, n, L, H, "first")
    rev_spa, util_spa = simulate_auction(trials, n, L, H, "second")
    revenues_fpa.append(rev_fpa.mean())
    revenues_spa.append(rev_spa.mean())
    utilities_fpa.append(util_fpa.mean())
    utilities_spa.append(util_spa.mean())


# Histogram of revenue distribution when n = 5
rev_fpa, _ = simulate_auction(trials, 5, L, H, "first")
rev_spa, _ = simulate_auction(trials, 5, L, H, "second")

plt.figure()
plt.hist(rev_fpa, bins=40, alpha=0.6, label="First-Price", edgecolor="black")
plt.hist(rev_spa, bins=40, alpha=0.6, label="Second-Price", edgecolor="black")
plt.title("Revenue Distribution (n = 5)")
plt.xlabel("Revenue")
plt.ylabel("Frequency")
plt.legend(frameon=False)
plt.tight_layout()
plt.show()


# Line plot mean revenue vs number of bidders
plt.figure()
plt.plot(n_values, revenues_fpa, marker="o", label="First-Price", linewidth=1.8)
```

```
plt.plot(n_values, revenues_spa, marker="o", label="Second-Price", linewidth=1.8)
plt.title("Mean Revenue vs. Number of Bidders")
plt.xlabel("Number of Bidders (n)")
plt.ylabel("Mean Revenue")
plt.legend(frameon=False)
plt.tight_layout()
plt.show()

# Line plot mean winner utility vs number of bidders
plt.figure()
plt.plot(n_values, utilities_fpa, marker="o", label="First-Price", linewidth=1.8)
plt.plot(n_values, utilities_spa, marker="o", label="Second-Price", linewidth=1.8)
plt.title("Mean Winner Utility vs. Number of Bidders")
plt.xlabel("Number of Bidders (n)")
plt.ylabel("Mean Winner Utility")
plt.legend(frameon=False)
plt.tight_layout()
plt.show()
```